

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I K Charan (192524488) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Charan

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

APPLICATION BASED QUESTION:

1. Auto-Scaling and Load Balancing for Seasonal Traffic

Introduction

An e-commerce company may experience unpredictable traffic during seasonal sales, festivals, and special offers. Cloud computing provides auto-scaling and load balancing to manage sudden increases in users without affecting application performance.

Auto-Scaling

Auto-scaling automatically increases or decreases the number of servers according to the workload. When traffic increases, additional servers are started automatically. When traffic decreases, unnecessary servers are removed. This provides flexibility and also reduces cloud infrastructure costs.

Load Balancing

Load balancing distributes incoming user requests among multiple servers. It prevents a single server from becoming overloaded. If one server fails, the load balancer redirects requests to healthy servers, improving system availability and reliability.

Role of Monitoring

Monitoring tools continuously observe CPU usage, memory, network traffic, response time, and server health. When resource usage becomes high, alerts can be generated and auto-scaling can be triggered. Monitoring also helps administrators identify failures and performance problems.

Real-World Example

During a major festival sale, an e-commerce website may receive millions of customer requests. Auto-scaling adds more servers, while load balancing distributes customers among them. This prevents crashes, reduces downtime, and provides faster page loading and smoother checkout.

Benefits

- Handles sudden traffic increases.
- Reduces downtime.
- Improves application performance.
- Provides better user experience.
- Reduces unnecessary infrastructure costs.

2. Secure Cloud Storage for Backup Data

Introduction

Organizations generate large amounts of important data and require reliable backup storage. Cloud storage provides scalable, durable, and secure storage without requiring organizations to maintain large physical storage infrastructure.

Data Durability

Cloud storage provides high durability by protecting stored data against hardware failures. Data is stored using reliable storage systems so that the possibility of permanent data loss is very low.

Data Redundancy

Redundancy means maintaining multiple copies of data. Copies can be stored across different storage devices or locations. If one copy becomes unavailable, another copy can be used for recovery.

Encryption

Encryption converts backup data into an unreadable format. Data can be encrypted both when stored in the cloud and when transferred through a network. This prevents unauthorized users from understanding sensitive backup information.

Access Control

Access control ensures that only authorized users can access backup data. Different permissions can be provided to administrators, employees, and backup operators. Multi-factor authentication can provide additional security.

Cost Advantages

Cloud storage reduces the need to purchase physical storage devices, backup servers, and data-center infrastructure. Organizations can increase or decrease storage based on their requirements and generally pay according to their usage.

Disaster Recovery Example

If a company's data center is damaged because of a fire, flood, or hardware failure, the organization can restore its applications and databases from cloud backups. This helps the organization resume its operations quickly.

Benefits

- Reliable backup storage.
- Protection against data loss.
- Strong security.
- Easy scalability.
- Lower infrastructure cost.
- Faster disaster recovery.

3. CDN and Edge Servers for Global Applications

Introduction

A mobile application with users in different countries must provide fast and reliable service. CDN and edge servers help deliver application content closer to users and reduce network latency.

Content Delivery Network

A **Content Delivery Network (CDN)** consists of multiple servers located in different geographical regions. Frequently accessed content such as images, videos, files, and application resources is cached on these servers.

Reducing Latency

When a user requests content, the CDN delivers it from a nearby server instead of the main server. Since the data travels a shorter distance, response time is reduced and application performance improves.

Edge Servers

Edge servers are located close to end users and can deliver content or perform certain processing near the user. This reduces the need to communicate with a distant central data center.

Global Availability

Cloud providers operate data centers in multiple regions and availability zones. Applications can be deployed across different regions so that users can access services from nearby locations. If one region fails, traffic can be redirected to another region.

Real-World Example

In a real-time gaming application, players from different countries need fast responses. CDN and edge servers can deliver game content from nearby locations, reducing delay and improving the gaming experience.

Benefits

- Reduces latency.
- Improves application speed.
- Reduces server workload.
- Supports global users.
- Improves availability.
- Provides a better user experience.

4. Cloud Security and Compliance for Financial Institutions

Introduction

Financial institutions handle sensitive information such as customer details, bank accounts, payment information, and transaction records. Therefore, strong security and regulatory compliance are essential when using cloud services.

Regulatory Compliance

Cloud providers offer security controls, compliance frameworks, auditing facilities, and identity management services. These features help financial organizations meet required security and regulatory standards.

Logging

Logging records important activities such as user login, database access, file modifications, administrative actions, and financial transactions. Logs are useful for auditing and investigating security incidents.

Monitoring

Monitoring continuously checks the cloud environment for unusual activities, system failures, and security threats. Alerts can be generated when suspicious activities are detected.

Encryption

Encryption protects sensitive financial data by converting it into an unreadable format. Data should be encrypted both when stored and while being transferred between systems.

Access Control

Identity and Access Management ensures that only authorized users can access sensitive information. Role-based permissions and multi-factor authentication can provide additional protection.

Data Sovereignty

Data sovereignty requirements can be satisfied by selecting appropriate cloud regions where the organization's data is legally allowed to be stored and processed. The organization should also follow applicable data protection regulations.

Example

When a customer transfers money through online banking, transaction information is encrypted and access is restricted to authorized systems. Logging records the transaction, while monitoring tools can detect unusual transactions.

Benefits

- Protects sensitive financial data.
- Supports regulatory compliance.
- Provides auditing capabilities.
- Detects security threats.
- Controls unauthorized access.

5. Containers and Kubernetes

Introduction

Containerization allows applications to be packaged with their required libraries and dependencies. It provides a consistent environment for developing, testing, and deploying applications.

Containers vs Virtual Machines

Containers are lightweight because they share the host operating system kernel. They require fewer resources and can start quickly. Virtual machines include a complete guest operating system, so they generally require more memory, storage, and processing power.

Kubernetes Orchestration

Kubernetes is a container orchestration platform used to manage multiple containers. It automatically deploys containers, monitors their health, restarts failed containers, and manages application resources.

Scaling

Kubernetes can increase or decrease the number of containers according to application demand. During high traffic, more containers can be created. When traffic decreases, unnecessary containers can be removed.

Portability

Containers can run consistently across different environments. The same container can be deployed on different cloud providers or on-premises infrastructure with minimal changes. This improves portability and reduces dependency on a single cloud platform.

Microservices Example

An online shopping application can be divided into different microservices such as:

- User Management
- Product Catalog
- Shopping Cart
- Payment
- Order Management
- Notification

Each service can run in a separate container. Kubernetes manages these containers and scales individual services according to their workload.